Harness



File: harness.kicad_sch

Daughterboard



File: daughter.kicad_sch

Wideband, MAP, USB, AUX



File: io.kicad_sch

Miata ECU motherboard 1989–1995 (NA6 / NA8). Carries the year-specific harness header and the year-independent I/O (wideband, MAP, IAT, USB-C). All engine control is on the daughterboard.

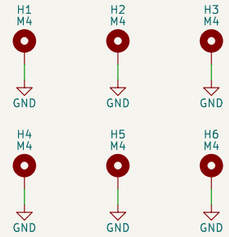
Sheet: /
File: miata_mb_8995.kicad_sch

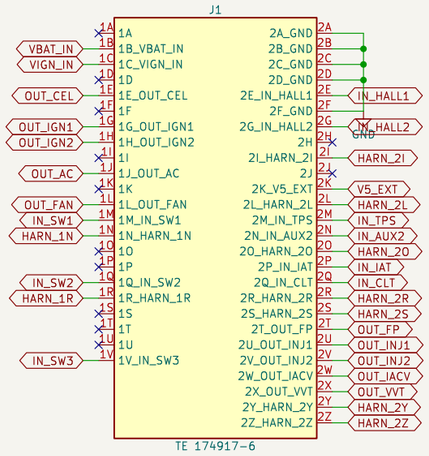
Title: Miata ECU motherboard 1989–1995 (NA6 / NA8)

Size: A4	Date: 2026-10-30	**Rev: 0.12**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

Stock harness header (Miata ECU motherboard 1989–1995): every pin labelled with the daughterboard signal it drives

Case mounting bosses (M4), tied to ground





Generated from scripts/mb_na6.py; netlist identical to the routed board

Sheet: /Harness/
File: harness.kicad_sch

Title: Miata ECU motherboard 1989–1995 — Harness

Size: A3	Date: 2026-10-30	**Rev: 0.12**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

Daughterboard sockets: pin-for-pin with the daughterboard J1 (top) and J2 (bottom)

Daughterboard standoffs (M3)

HD1 M3 standoff
HD2 M3 standoff
HD3 M3 standoff
HD4 M3 standoff



Generated from scripts/mb_na6.py; netlist identical to the routed board

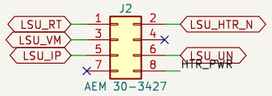
Sheet: /Daughterboard/
File: daughter.kicad_sch

Title: Miata ECU motherboard 1989–1995 — Daughterboard

Size: A3	Date: 2026-10-30	**Rev: 0.12**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/4

Wideband sensor harness AEM 30-3427 (controller on the daughterboard)

AEM pin: 1 trim (RT) 2 H− 3 common (VM) 5 pump (IP) 6 sense (UN) 8 H+. Micro-Fit 2x4 ASSUMED

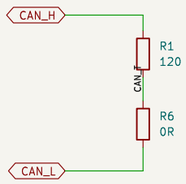




Heater fuse: 2 A hold, 30 V PTC

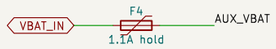
IP/VM/RT/UN and H− run to daughterboard J2 33–38; the heater current returns on J2 39/40

CAN bus termination at the ECU end

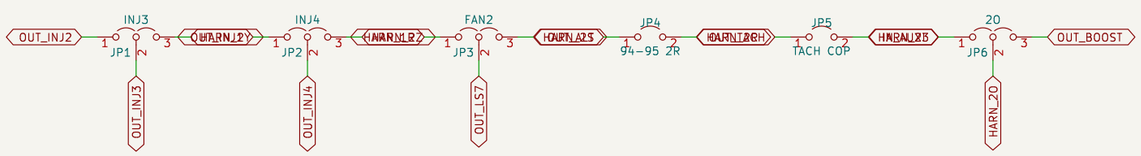


120R; remove R6 if the bus is terminated elsewhere

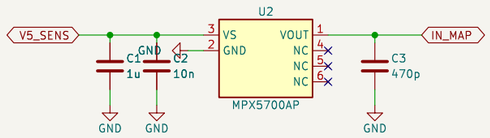
Fused +12 V feeds (harness actuators, AUX)



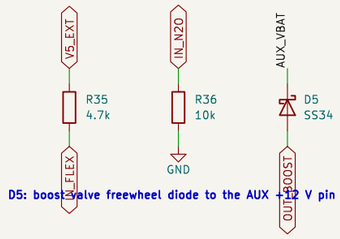
1989–95 year / option jumpers, injector clamps (1989–93 paired wires), tachometer pull-up (JP5)



MAP sensor

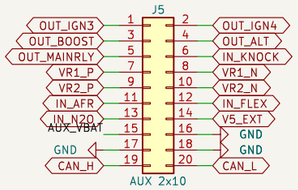


Flex-fuel sensor pull-up (5 V) and nitrous arming pull-down; both inputs are on the AUX header

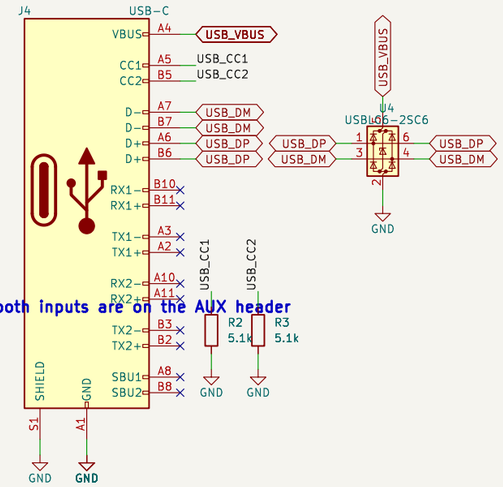


D5: boost-valve freewheel diode to the AUX +12 V pin

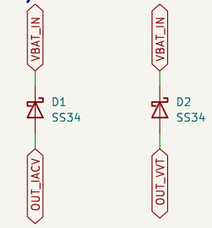
AUX: add-on I/O the stock harness has no pins for (+12 V through F4)



USB-C (device): tuning, logs, DFU firmware update

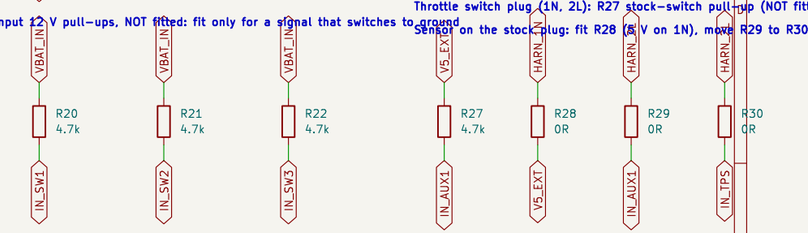


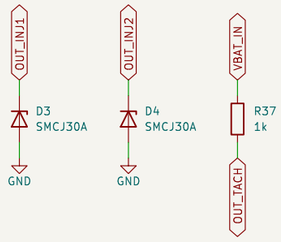
Valve freewheel diodes (to each valve's +12 V feed) and fit options



Switch-input 12 V pull-ups, NOT fitted: fit only for a signal that switches to ground

Throttle switch plug (1N, 2L): R27 stock-switch pull-up (NOT fitted); R28 sensor on the stock plug: fit R28 (5 V on 1N), move R29 to R30





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File: io.kicad_sch

Title: Miata ECU motherboard 1989–1995 – Wideband, MAP, USB, AUX

Size: A3	Date: 2026-10-30	**Rev: 0.12**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/4